DO NOW — RECALL

# Lesson 060 · Refining an Abstraction and Testing Assum

Name: _____ Class: _____ Date: _____

**1. What does abstraction remove?**

**2. What is a model assumption?**

**3. What is a boundary test?**

**4. Two rising-edge D-type flip-flops A and B share a clock and start with QA=0 and QB=0. DA=1 throughout, and DB is connected to QA. Explain why QB is still 0 immediately after the first rising edge. Assume ideal simultaneous sampling.**

DO NOW — CHECK / OBJECTIVES

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

Irrelevant details.

### 2. Model answer

Simplifying rule treated as true for a task.

### 3. Model answer

An input at/around an important threshold.

### 4. Model answer

Both sample their inputs at the same edge. B samples the old QA=0, while A samples 1. A's new output reaches B after that sampling event.

### By the end, you can…

• Compare the simplification against new requirements. If the model omits a relevant factor it can produce plausible but wrong outputs.

• Refine the abstraction by adding only the newly necessary variable, rather than reconstructing every real-world detail.

• Test normal, boundary and exceptional cases: 5 km versus 6 km, return and single, and invalid negative distance.

### Vocabulary — know these terms

refinement — Improving model for new requirements.

boundary — Important threshold or edge.

validity — Whether model works for its intended use.

assumption — Rule taken as given.

NEW LEARNING

# Read, annotate and explain

### Understand

Compare the simplification against new requirements. If the model omits a relevant factor it can produce plausible but wrong outputs.

### Model

Refine the abstraction by adding only the newly necessary variable, rather than reconstructing every real-world detail.

### Evaluate

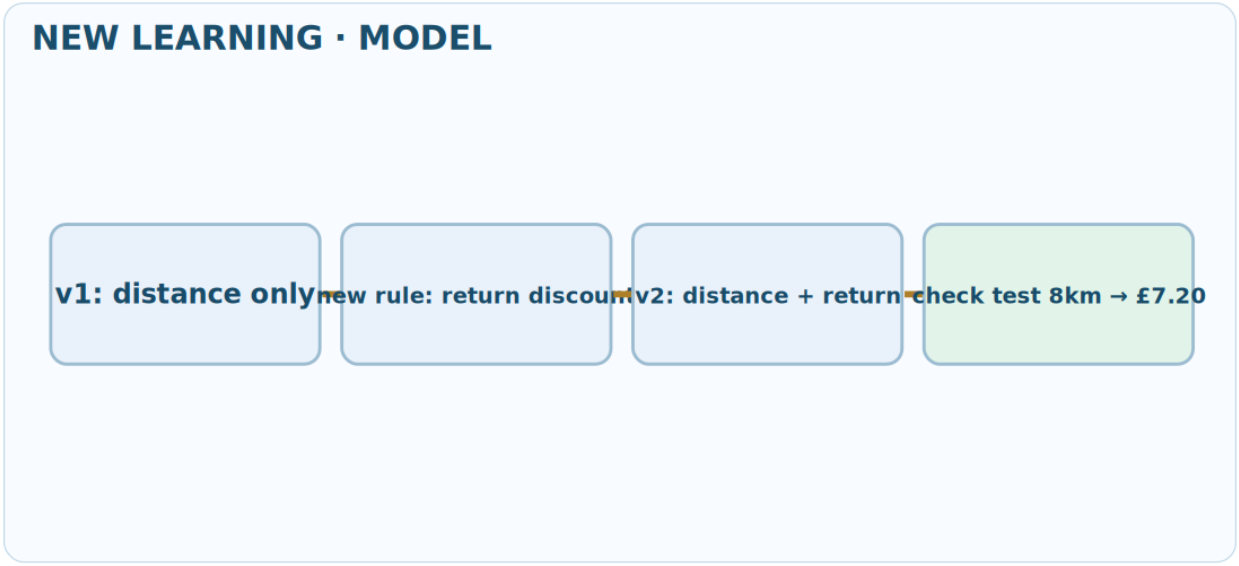
Test normal, boundary and exceptional cases: 5 km versus 6 km, return and single, and invalid negative distance.

**Explain one step in your own words, using the correct technical terms.**

# New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### New Learning




NEW LEARNING · MODEL: v1: distance only; new rule: return discount; v2: distance + return; check test 8km → £7.20

I DO — WORKED EXAMPLE

# Follow the worked method

### A simplified cinema ticket model uses only age but now weekend prices differ. What change is needed?

### Explain

Weekend becomes a relevant distinction because it can change the output.

# I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**A simplified cinema ticket model uses only age but now weekend prices differ. What change is needed?**

Weekend becomes a relevant distinction because it can change the output.




IDO · TRACE: age; weekend; fare

WE DO — GUIDED PRACTICE

# Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

### Guided 1 — TRY

A parking price model charges £3/hour but introduces £12 maximum. Improve algorithm.
Source code / input:
fee=MIN(3×hours,12), with validation that hours≥0.

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

### Question 1

Ticket v1 charges £2 at ≤5km, £4 otherwise. Give charge at 5km and 6km and why these are useful tests. [6]

### Question 2

Return journeys cost 1.8 times single. For 8km journey calculate return fare and show variables. [6]

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

### Question 3

A school model predicts homework concerns from just missed work counts; discuss which relevant conditions to add and how to test impact. [6]

EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking.

**A model that worked yesterday never needs changing when requirements change.**

**Circle: Confident / Getting there / Need help**

Explain what you can now do independently.

## GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

WE DO — CHECK AND IMPROVE

# Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

## Guided 1 — CHECK

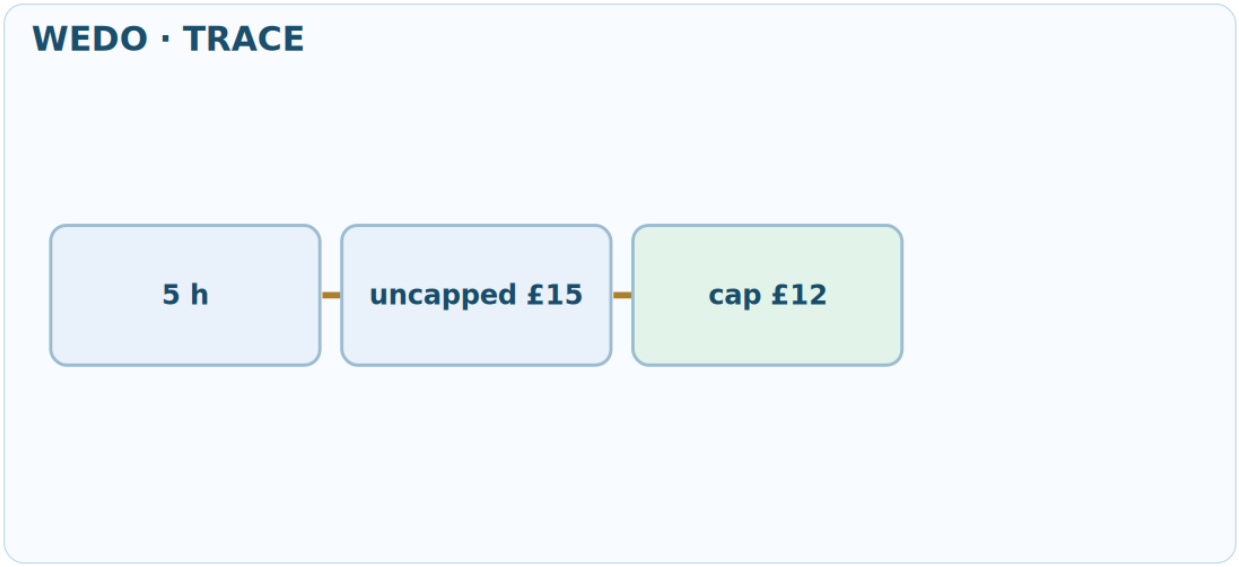
Explain
Cap is new relevant rule. Example 5 hours costs £12, not £15.

# We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**A parking price model charges £3/hour but introduces £12 maximum. Improve algorithm.**

Cap is new relevant rule. Example 5 hours costs £12, not £15.




WEDO · TRACE: 5 h; uncapped £15; cap £12